INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

==

THE SITE

Location	WI facility WI_way_1510420026 -- station KENW, America/Chicago
Building	OSM 1510420026 OSM way 1510420026
Facade gap	not applicable -- one building, no second facade
Weather record	41,755 real hourly records from KENW
Plume physics	NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-03-24 (clear_cool)
Plant limit	18.0 C
Notice required	6 h
Forecast skill	0.00 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 9 of 24 hours with 2 mode changes. 8 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	9 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	9 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s) 8 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

----

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	11.120	18.0	10.560	3.880
01	FREE	yes	-	11.120	18.0	10.000	3.165
02	FREE	yes	-	11.670	18.0	10.000	3.250
03	FREE	yes	-	11.670	18.0	10.560	2.961
04	FREE	yes	-	11.670	18.0	10.560	2.721
05	FREE	yes	-	11.670	18.0	10.560	2.522
06	FREE	yes	-	12.780	18.0	10.000	3.502
07	FREE	yes	-	14.440	18.0	11.670	4.251
08	FREE	yes	-	17.220	18.0	13.330	5.186

09	mechanical	no	dry-bulb	20.010	18.0	13.330	5.770
10	mechanical	no	dry-bulb	21.670	18.0	12.780	6.126
11	mechanical	no	dry-bulb	22.780	18.0	11.670	6.347
12	mechanical	no	dry-bulb	22.220	18.0	11.670	6.017
13	mechanical	no	dry-bulb	22.170	18.0	12.780	5.225
14	mechanical	no	dry-bulb	21.110	18.0	12.780	3.957
15	mechanical	no	dry-bulb	18.890	18.0	12.780	3.301
16	mechanical	yes	switch budget	16.120	18.0	13.890	2.700
17	mechanical	yes	switch budget	13.670	18.0	13.330	2.855
18	mechanical	yes	switch budget	12.230	18.0	12.220	2.834
19	mechanical	yes	switch budget	12.780	18.0	10.560	3.513
20	mechanical	yes	switch budget	12.780	18.0	9.440	4.266
21	mechanical	yes	switch budget	12.780	18.0	8.890	4.488
22	mechanical	yes	switch budget	13.890	18.0	8.330	4.355
23	mechanical	yes	switch budget	13.830	18.0	8.330	4.412

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.120 C, which is 6.880 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.880 C, and the margin is measured, not chosen: 3.880 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.120 C, which is 6.880 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.165 C, and the margin is measured, not chosen: 3.165 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.670 C, which is 6.330 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.250 C, and the margin is measured, not chosen: 3.250 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.670 C, which is 6.330 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.961 C, and the margin is measured, not chosen: 2.961 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.670 C, which is 6.330 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.722 C, and the margin is measured, not chosen: 2.722 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.670 C, which is 6.330 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.522 C, and the margin is measured, not chosen: 2.522 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.780 C, which is 5.220 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.502 C, and the margin is measured, not chosen: 3.502 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.440 C, which is 3.560 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.251 C, and the margin is measured, not chosen: 4.251 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.220 C, which is 0.780 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 5.186 C, and the margin is measured, not chosen: 5.186 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.010 C against a 18.0 C limit, so it fails by 2.010 C. It would take a limit 2.010 C higher, or a bound 2.010 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.670 C against a 18.0 C limit, so it fails by 3.670 C. It would take a limit 3.670 C higher, or a bound 3.670 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.780 C against a 18.0 C limit, so it fails by 4.780 C. It would take a limit 4.780 C higher, or a bound 4.780 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.220 C against a 18.0 C limit, so it fails by 4.220 C. It would take a limit 4.220 C higher, or a bound 4.220 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.170 C against a 18.0 C limit, so it fails by 4.170 C. It would take a limit 4.170 C higher, or a bound 4.170 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.110 C against a 18.0 C limit, so it fails by 3.110 C. It would take a limit 3.110 C higher, or a bound 3.110 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.890 C against a 18.0 C limit, so it fails by 0.890 C. It would take a limit 0.890 C higher, or a bound 0.890 C tighter, to change this hour.

16:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.120 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

17:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.670 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

18:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.230 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

19:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.780 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

20:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.780 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

21:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.780 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

22:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.890 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.830 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

Generated by INTAKE-ARBITER/src/report.py. Verified by being read back: the file was reopened after writing and every hour of the schedule, the headline counts and this site's own name were confirmed present.